

Equations of circles



1 Graph B

2

a No solution provided

b Quadrilateral

c Circle

3

a

i $(0,0)$

ii 2 units

b

i $(3,2)$

ii 3 units

c

i $(1,-2)$

ii 8 units

d

i $(-3,-1)$

ii 5 units

4 Every point on the circle is exactly 5 units away from the point $(0,0)$.

5

a 1

b 0

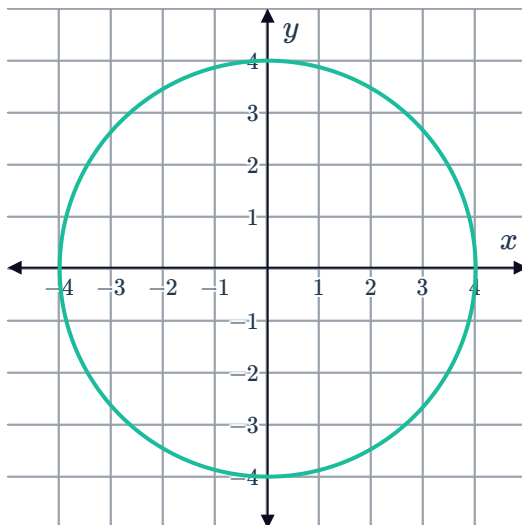
6 $x^2 + y^2 = 49$

7

a

i

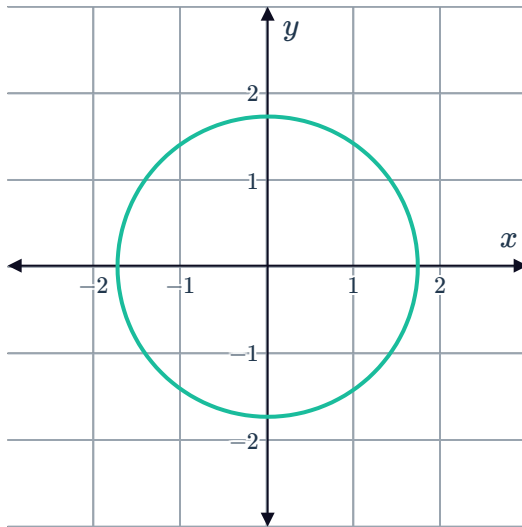
ii $x^2 + y^2 = 16$



b
i

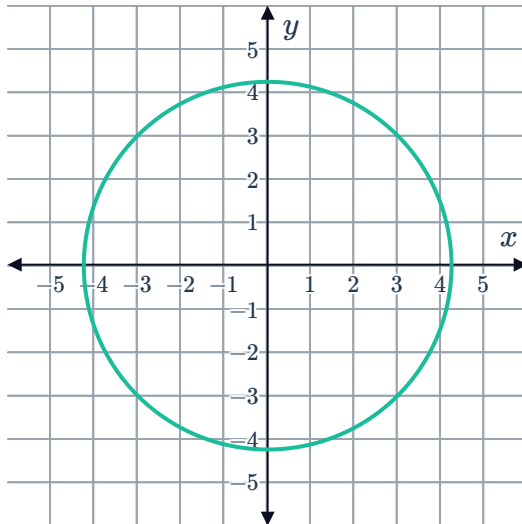


ii $x^2 + y^2 = 3$



c
i

ii $x^2 + y^2 = 18$



8

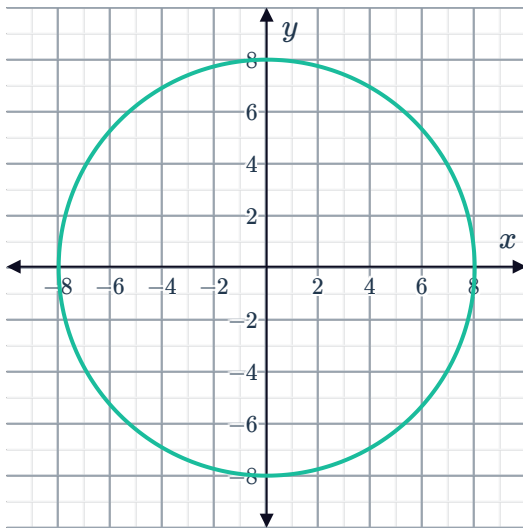
a $(0,0)$

c $x = 8, -8$

b 8 units

d $y = 8, -8$

e



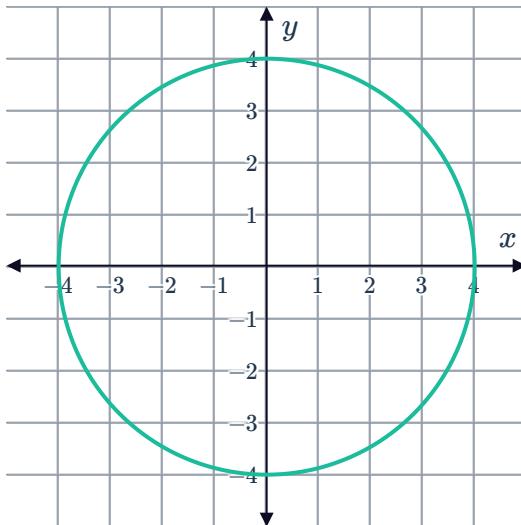
9 $4\sqrt{2}$ units

10

a

i 4 units

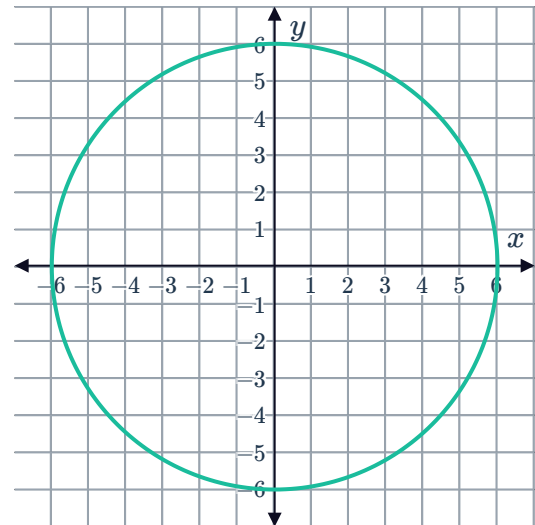
ii



b

i 6 units

ii



11 $x^2 + y^2 = 4$

12

a $5\sqrt{5}$ units

b $x^2 + y^2 = 125$

13

a $x^2 + y^2 = 36$

b 36π units²

14 a 14 units



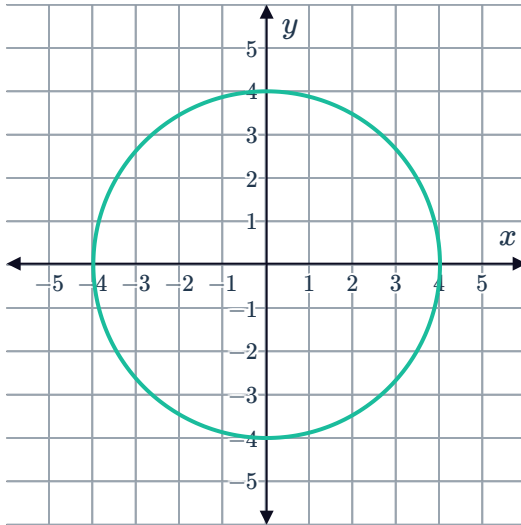
$$y = \pm\sqrt{48}$$

15 Inside

16

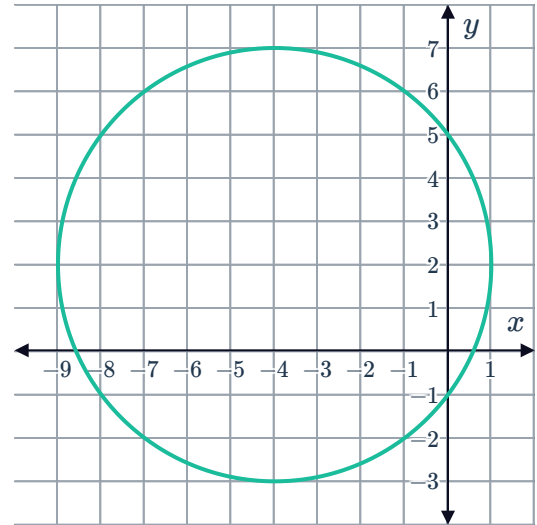
a

- i (0,0)
- ii 4 units
- iii



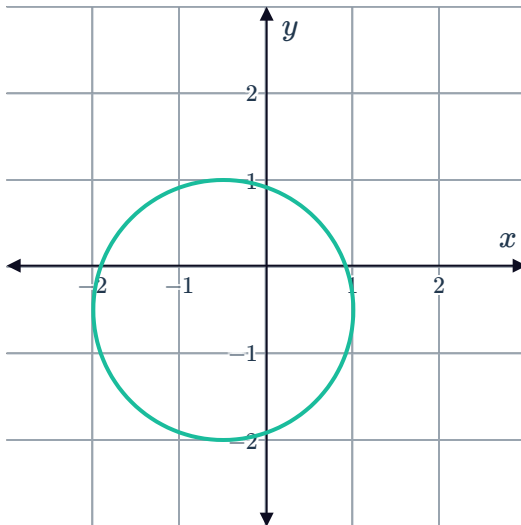
b

- i (-4,2)
- ii 5 units
- iii



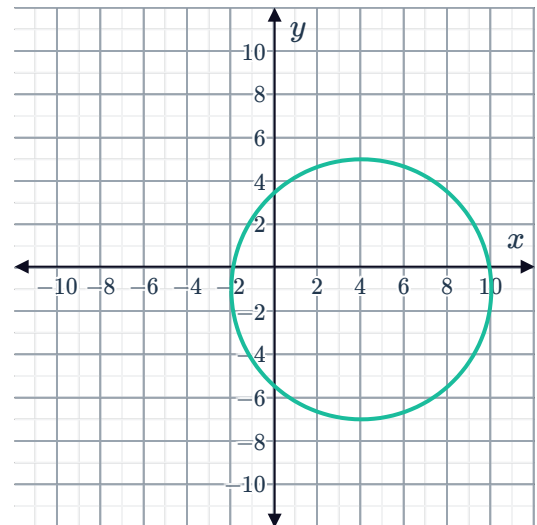
c

- i $\left(-\frac{1}{2}, -\frac{1}{2}\right)$
- ii $\frac{3}{2}$ units
- iii



d

- i (4,-1)
- ii 6 units
- iii

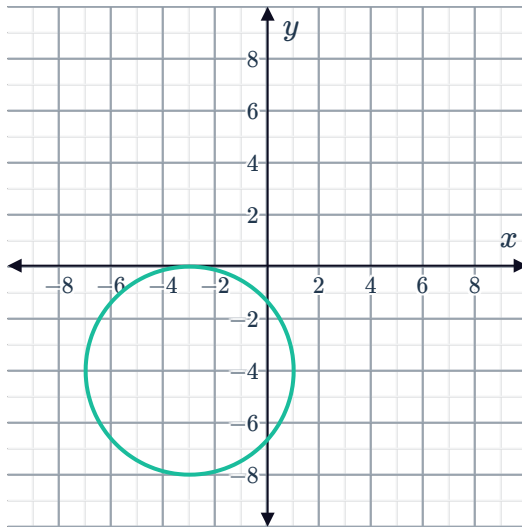


e

i $(-3, -4)$

ii 4 units

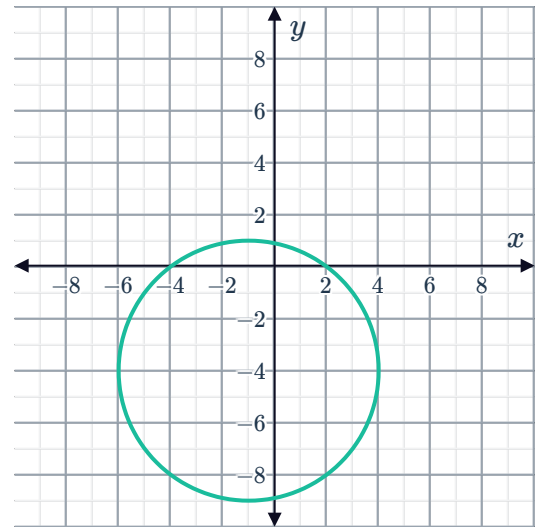
iii



i $(-1, -4)$

ii 5 units

iii

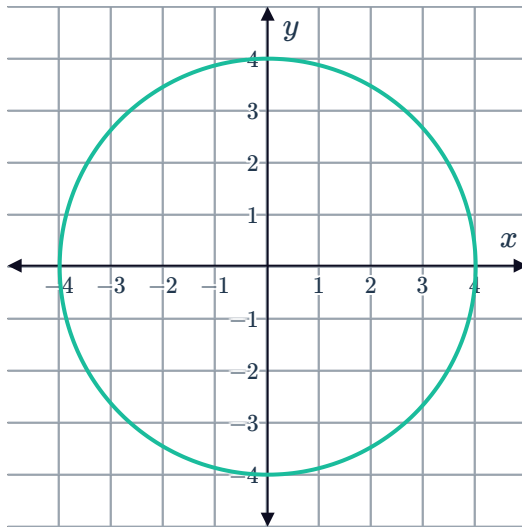


g

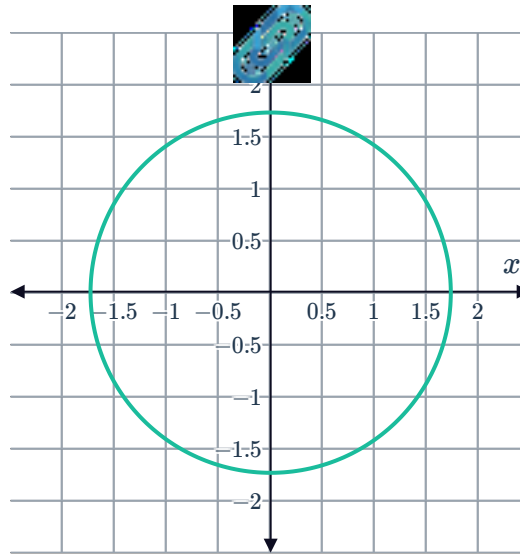
i $(0, 0)$

ii 4 units

iii



17



18 $\sqrt{353}$ units

19

a
i $\left(-3.1, \frac{3}{2}\right)$ ii $\frac{3}{4}$ units

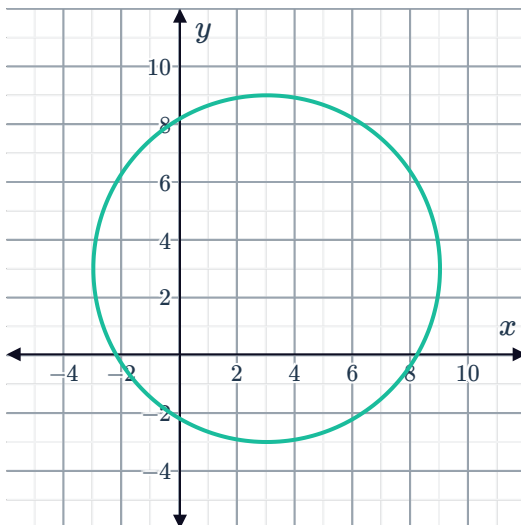
b
i $(6, 6)$ ii $2\sqrt{3}$ units

c
i $(0.7, -4.4)$ ii $\sqrt{3}$ units

d
i $(8, -1)$ ii $\sqrt{13}$ units

20

a



b $(x - 3)^2 + (y - 3)^2 = 36$

21

a

i $(-3, -3)$

ii 3 units

iii $(x + 3)^2 + (y + 3)^2 = 9$



b

i $(1, 3)$

ii 6 units

iii $(x - 1)^2 + (y - 3)^2 = 36$

22

a $(x - 1)^2 + (y + 3)^2 = 25$

b $(x + 5.7)^2 + y^2 = 9$

23 Two

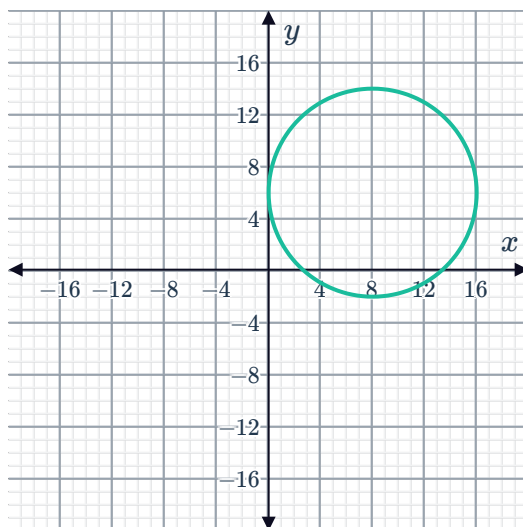
24

a $(x - 8)^2 + (y - 6)^2 = 64$

b No

25

a



b $x^2 - 4x + y^2 + 12y + 31 = 0$

26

a Outside

b On

c Inside

27

a Inside

b On

c On

Additional questions

1. A circle is centered at $(-3, -3)$ with a radius of 3 units. Write the equation of the circle.

29

a

- i (0,0)
- ii 5 units
- iii 10 units
- iv $x^2 + y^2 = 25$

b

- i (1, -2)
- ii 4 units
- iii 8 units
- iv $(x - 1)^2 + (y + 2)^2 = 16$



30

a (2, -3)

b $r = 5\sqrt{3}$ units

c $r = 10\sqrt{3}$ units

31

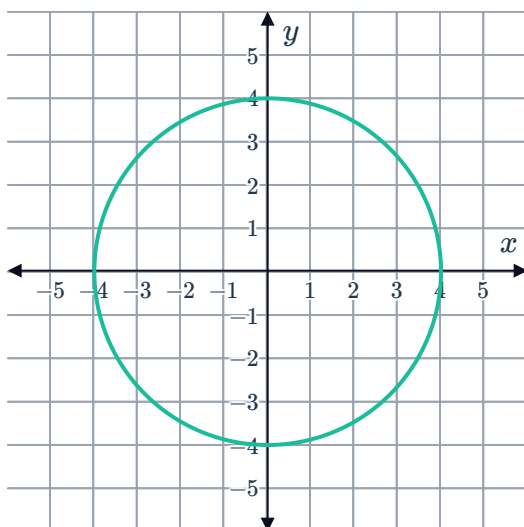
a $x^2 + (y - 6)^2 = 9$

b $\left(x - \frac{17}{2}\right)^2 + y^2 = 15$

32

a

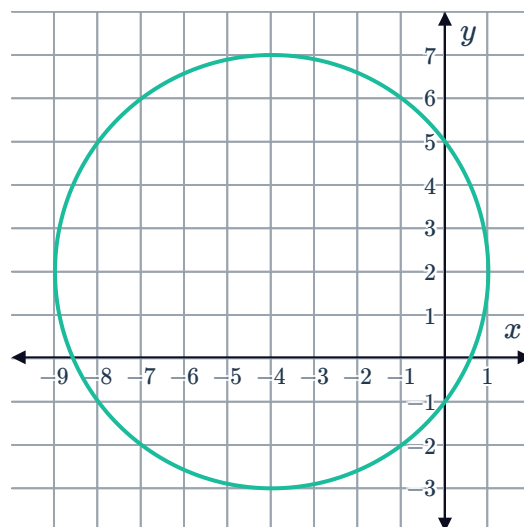
- i (0,0)
- ii 4 units
- iii



iv Domain = $[-4, 4]$
Range = $[-4, 4]$

b

- i (-4,2)
- ii 5 units
- iii



iv Domain = $[-9, 1]$
Range = $[-3, 7]$

33

a 13 units

b $(x + 2)^2 + (y - 5)^2 = 169$

34

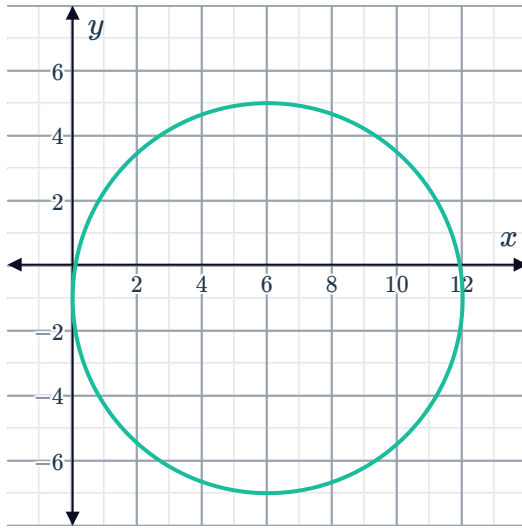


35

a

i $(x - 6)^2 + (y + 1)^2 = 36$

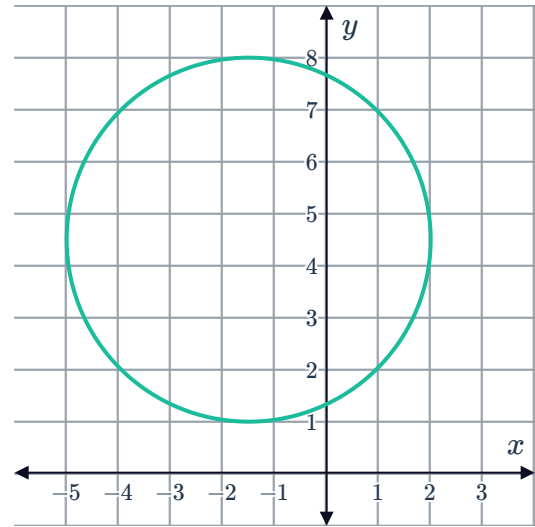
ii



b

i $\left(x + \frac{3}{2}\right)^2 + \left(y - \frac{9}{2}\right)^2 = \frac{49}{4}$

ii



36 $(x - 25)^2 + (y - 30)^2 = 400$

37

a $(49.5, 37.5)$

c $(x - 49.5)^2 + (y - 37.5)^2 = 83.7225$

b 9.15 m

d Domain = $[40.35, 58.65]$
Range = $[28.35, 46.65]$

38

a Domain: $[5, 19]$, Range: $[1, 15]$

c 5 cm

b Domain: $[14, 26]$, Range: $[2, 14]$

d 85.19 cm^2

39



a $(x - 4)^2 + (y + 3)^2 = 41$

b Given:

- Equation of circle: $(x - 4)^2 + (y + 3)^2 = 41$
- Point: $P(8, 3)$

To prove: Distance between circle center and $P(8, 3)$ is greater than $\sqrt{41}$

1. Find the center of the circle

$$(x - 4)^2 + (y + 3)^2 = 41 \quad \text{Given}$$

$$C = (4, -3) \quad \text{Equation of the circle}$$

2. Find distance between center and $C(8, 3)$

$$CP = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad \text{Distance formula}$$

$$= \sqrt{(8 - 4)^2 + (3 - -3)^2} \quad \text{Substitution}$$

$$= \sqrt{(4)^2 + (6)^2} \quad \text{Simplifying}$$

$$= \sqrt{16 + 36} \quad \text{Simplifying}$$

$$= \sqrt{52} \quad \text{Simplifying}$$

3. Show distance between circle center and $P(8, 3)$ is greater than $\sqrt{41}$

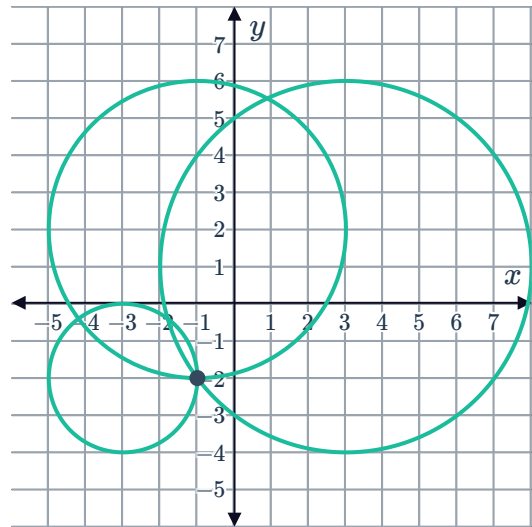
$$CP = \sqrt{52} > \sqrt{41}$$

Therefore, P is outside the circle.

40



- a As the point of intersection of the three circles centered at the receiving stations with radii as the distances, the epicenter is at $(-1, -2)$.



- b The point $(8, -7)$ is outside the circle with center $(-1, -2)$ with radius 10, so it would not feel the earthquake.